Quiz 2

Question 1 (10 Points)

Watchlist (**streamerID**, **showName**, **SeasonNo**, streamerName, subscriptionPlan, region, genre, releaseYear, totalEpisodes, producerPhone, producerName, producerEmail, platform, TRP, watchTimeLine, userRating)

The primary key is underlined.

The relation has the following additional functional dependencies (FDs):

FD1: streamerID → streamerName, subscriptionPlan, region, platform

FD2: showName → genre, producerName, producerPhone, producerEmail, releaseYear

FD3: producerName → producerPhone, producerEmail

FD4: showName, SeasonNo → TRP

FD5: subscriptionPlan, region → platform

a. Explain if this is in **1NF** or not. If not, decompose it to 1NF. [2]

b. Explain if this is in **2NF** or not. If not, decompose it to 2NF. [4]

c . Explain if this is in **3NF** or not. If not, decompose it to 3NF. [4]

Solution:

a. Not in 1NF due to no composite, multivalued attribute and no nested relation
b. Not in 2NF due to partial FD of FD1, FD2, FD4

   Decomposing to 2NF:

   Streamer
   (streamerID, streamerName, subscriptionPlan, region, platform)
   Shows
   (showName, genre, producerName, producerPhone, producerEmail, releaseYear)
   Rating
   (showName, SeasonNo , TRP)
   Watchlist
   (streamerID, showName, SeasonNo, totalEpisodes, watchTimeLine, userRating)

c. Not in 3NF due to transitive dependency of FD3, FD5

   Decomposing to 3NF:

   Shows
   (showName, genre, producerName, releaseYear)
   Producer
   (producerName, producerPhone, producerEmail)

   Streamer
   (streamerID, streamerName, subscriptionPlan, region)

Plans
(subscriptionPlan, region, platform)

Rating and Watchlist relation remains unchanged.